

Not All Rollouts are Useful: Down-Sampling Rollouts in LLM Reinforcement Learning

并非所有 rollout 都有用：在 LLM 强化学习中下采样 rollout

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Abstract

摘要

Reinforcement learning with verifiable rewards (RLVR) has emerged as the leading approach for enhancing reasoning capabilities in large language models. However, it faces a fundamental compute and memory asymmetry: rollout generation is embarrassingly parallel and memory-light, whereas policy updates are communication-heavy and memory-intensive. To address this, we introduce PODS (Policy Optimization with Down-Sampling), which decouples rollout generation from policy updates by training only on a strategically selected subset of rollouts, maintaining learning quality while dramatically reducing update costs. We propose a principled subset selection criterion—max-variance down-sampling—that maximizes the variance of reward in the selected subset, and provide an efficient $O(n \log n)$ implementation of this rule. Empirically, Group Relative Policy Optimization (GRPO) coupled with PODS achieves the peak test accuracy of vanilla GRPO at least $1.7 \times$ faster across the different reasoning benchmarks and hardware configurations we tested.

基于可验证奖励的强化学习（RLVR）已成为增强大型语言模型推理能力的主导方法。然而，它面临一个根本性的计算和内存不对称问题：rollout 生成是 embarrassingly parallel（易并行）且内存轻量的，而策略更新则通信密集且内存消耗大。为了解决这个问题，我们引入了 PODS（基于下采样的策略优化），通过仅在策略更新中使用精心选择的 rollout 子集进行训练，将 rollout 生成与策略更新解耦，从而在保持学习质量的同时大幅降低更新成本。我们提出了一种基于最大方差下采样的子集选择准则，该准则最大化所选子集中奖励的方差，并提供了这一规则的高效 $O(n \log n)$ 实现。实验证明，在我们测试的不同推理基准和硬件配置下，结合 PODS 的组相对策略优化（GRPO）在至少 $1.7 \times$ faster 的情况下达到了与原始 GRPO 相同的峰值测试准确率。

1 Introduction

1 引言

Reinforcement learning with verifiable rewards (RLVR) has driven recent breakthroughs in solving math problems, code generation, and general reasoning for large language models (LLMs) (Jaech et al., 2024; Ziegler et al., 2019; Ouyang et al., 2022; Stiennon et al., 2020). RLVR algorithms such as Proximal Policy Optimization (PPO) (Schulman et al., 2017) and Group Relative Policy Optimization (GRPO) (Shao et al., 2024) share a two-phase structure: an inference phase, which generates rollouts given a prompt, and a policy-update phase, which updates the model parameters using the rewards calculated on those rollouts.

以可验证奖励（RLVR）进行强化学习已推动了大型语言模型（LLMs）在解决数学问题、代码生成和一般推理方面的近期突破（Jaech 等，2024；Ziegler 等，2019；Ouyang 等，2022；Stiennon 等，2020）。诸如近端策略优化（PPO）（Schulman 等，2017）和群体相对策略优化（GRPO）（Shao 等，2024）等 RLVR 算法共享一个两阶段结构：一个推理阶段，根据提示生成轨迹；以及一个策略更新阶段，利用在这些轨迹上计算出的奖励来更新模型参数。

These two phases place different computational demands on the hardware. Inference is embarrassingly parallel and relatively memory-light, enabling modern accelerators to produce thousands of rollouts concurrently. Although generating a single rollout may have high latency due to autoregressive decoding, batching rollouts amortizes the per-token latency and yields higher throughput. Policy updates, on the other hand, scale poorly with batch size: they are memory- and communication-intensive, requiring full-precision optimizer states and cross-device synchronization of gradients and parameters. This asymmetry creates a fundamental bottleneck: systems must either throttle inference (underutilizing compute) or resort to memory-saving techniques like gradient accumulation (increasing communication overhead and policy update latency), both of which hurt training efficiency. Fig. 1 provides empirical evidence for this computational asymmetry.

这两个阶段对硬件的计算需求不同。推理（Inference）是高度并行的，并且对内存的需求较低，使得现代加速器可以同时生成数千个轨迹（rollouts）。尽管生成单个轨迹可能由于自回归解码而具有较高的延迟，但通过批量处理轨迹可以摊薄每个 token 的延迟，从而提高吞吐量。另一方面，策略更新（Policy updates）在批量大小增加时扩展性较差：它们对内存和通信需求很高，需要全精度的优化器状态，并且需要在不同设备之间同步梯度和参数。这种不对称性创造了一个根本性的瓶颈：系统必须在限制推理（未充分利用计算资源）和采用节省内存的技术（如梯度累积，增加通信开销和策略更新延迟）之间做出选择，这两种方法都会损害训练效率。图 1 提供了这种计算不对称性的实证证据。

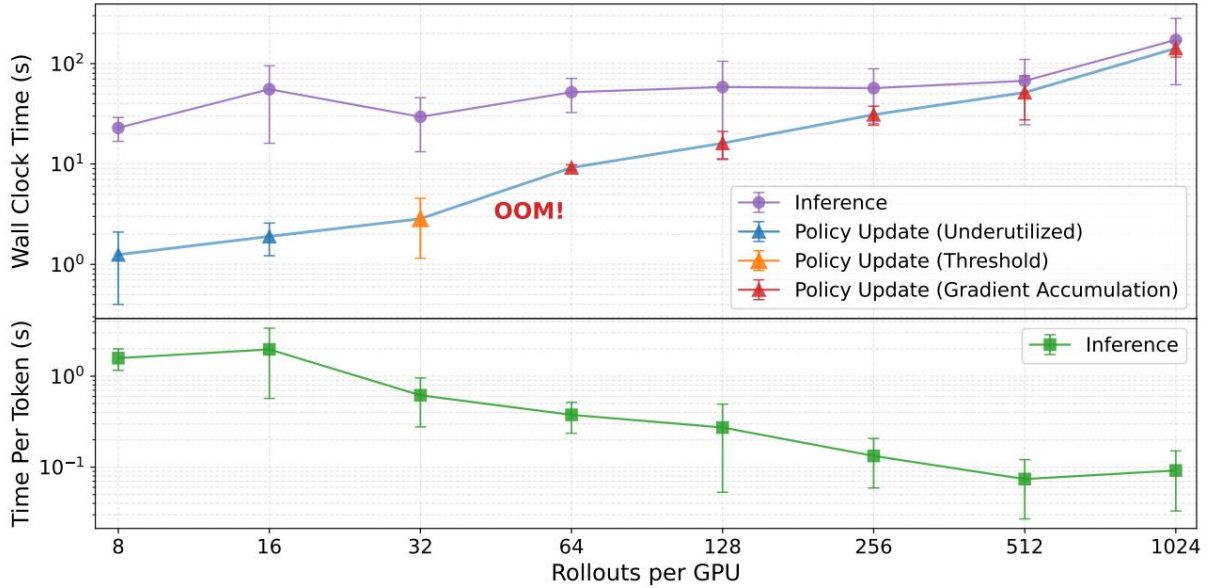


Figure 1: images/d921fba963af8e010294f7c18ceb9f968598a76ee238f53d6b9431619775fe9a.jpg

Figure 1: Inference scales efficiently while policy updates become memory-bound in RLVR. Empirical timing breakdown when fine-tuning Qwen2.5-3B-Instruct on GSM8K using 8 A100-80GB GPUs with varying rollouts per GPU. Top: Total wall-clock time per iteration. Policy updates hit memory limits after 32 rollouts per GPU (Out of memory beyond this point), requiring gradient accumulation that dramatically slows training. Bottom: Per-token inference time decreases $21\times$ through batching (from 8 to 512 rollouts), saturating beyond 512. This demonstrates the core asymmetry that PODS exploits: inference parallelizes efficiently while policy updates become memory-bound.

图 1：在 RLVR 中，推理可以高效扩展，而策略更新则变得受限于内存。在使用 8 块 A100-80GB GPU、每块 GPU 使用不同数量的 rollouts 对 Qwen2.5-3B-Instruct 进行微调时，实测时间分解。顶部：每次迭代的总实际时间。策略更新在每块 GPU 进行 32 次 rollouts 后达到内存限制（在此点之后出现内存不足），需要梯度累积，这会显著减缓训练速度。底部：每 token 的推理时间通过批处理（从 8 次到 512 次 rollouts）减少 $21\times$ ，在超过 512 次后趋于饱和。这展示了 PODS 所利用的核心不对称性：推理可以高效并行化，而策略更新则受限于内存。

A key observation of our work is: not all rollouts contribute equally to model improvement. Beyond a certain scale, additional rollouts provide diminishing returns and can even degrade learning signals through redundant information. This suggests a natural solution: generate large batches of rollouts during the scalable inference phase, but train selectively on only the most informative subset during the policy update phase, avoiding the latency overhead of memory-saving techniques. We formalize this idea in PODS (Policy Optimization with Down-Sampling). As illustrated in Fig. 2, PODS maximizes hardware utilization by generating n rollouts per prompt but updating on only $m < n$ informative samples selected by a principled down-sampling rule.

我们工作的一个关键观察是：并非所有 rollout 对模型改进的贡献是相同的。在一定规模之后，额外的 rollout 会带来递减的回报，甚至可能通过冗余信息损害学习信号。这提示了一个自然的解决方案：在可

扩展推理阶段生成大量 rollout，但在策略更新阶段仅选择最有信息量的子集进行训练，从而避免节省内存技术带来的延迟开销。我们在 PODS（基于下采样的策略优化）中正式提出了这一想法。如图 2 所示，PODS 通过每个提示生成 n 个 rollout，但仅根据一个原理性的下采样规则选择 $m < n$ 个有信息量的样本进行更新，以最大化硬件利用率。

Within the PODS framework, we introduce max-variance down-sampling, a principled criterion that selects the subset of rollouts with the greatest reward variance of the selected subset, thereby preserving strong contrastive signals. We show that the resulting combinatorial problem can be solved in $O(n \log n)$ time and, in the common binary-reward setting, reduces to picking the $m/2$ highest-reward and $m/2$ lowest-reward rollouts. We evaluate PODS with GRPO on GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021) and the Chemistry subset of SciKnowEval (Feng et al., 2024) across multiple model and hardware configurations, demonstrating that it achieves the peak test accuracy of baseline GRPO at least $1.7 \times$ faster.

在 PODS 框架中，我们引入了最大方差下采样（max-variance down-sampling），这是一种有原则的标准，通过选择具有最大奖励方差的 rollout 子集，从而保留强对比信号。我们展示了由此产生的组合问题可以在 $O(n \log n)$ 时间内解决，并且在常见的二元奖励设置下，简化为选择 $m/2$ 个最高奖励和 $m/2$ 个最低奖励的 rollout。我们在多种模型和硬件配置下，使用 GRPO 对 PODS 在 GSM8K (Cobbe 等人, 2021)、MATH (Hendrycks 等人, 2021) 以及 SciKnowEval 的化学子集 (Feng 等人, 2024) 上进行了评估，证明其至少达到基线 GRPO 的峰值测试准确率 $1.7 \times$ faster。

2 Related Work

2 相关工作

Reinforcement learning for LLM reasoning. Reinforcement learning has emerged as a powerful paradigm for enhancing the reasoning capabilities of LLMs across math, coding, and problem-solving domains (Jaech et al., 2024; Shao et al., 2024; Kazemnejad et al., 2024). Although classical algorithms such as Proximal Policy Optimization (PPO) (Schulman et al., 2017) laid the foundation, recent work has tailored them specifically to language models. Group Relative Policy Optimization (GRPO) (Shao et al., 2024) has gained prominence for reasoning tasks because of its implementation simplicity, competitive performance relative to PPO, and lack of a separate critic network. OpenAI o1 (OpenAI, 2024) and DeepSeek R1 (Guo et al., 2025),

强化学习用于大语言模型 (LLM) 推理。强化学习作为一种强大的范式，已被用于增强 LLM 在数学、编程和问题解决等领域的推理能力 (Jaech 等, 2024; Shao 等, 2024; Kazemnejad 等, 2024)。尽管诸如近端策略优化 (PPO) (Schulman 等, 2017) 等经典算法奠定了基础，但近期的研究工作则专门针对语言模型进行了调整。基于组相对策略优化 (GRPO) (Shao 等, 2024) 因其实现简便、相对于 PPO 具有竞争力的性能以及不需要单独的批评网络而受到推理任务的青睐。OpenAI o1 (OpenAI, 2024) 和 DeepSeek R1 (Guo 等, 2025)，

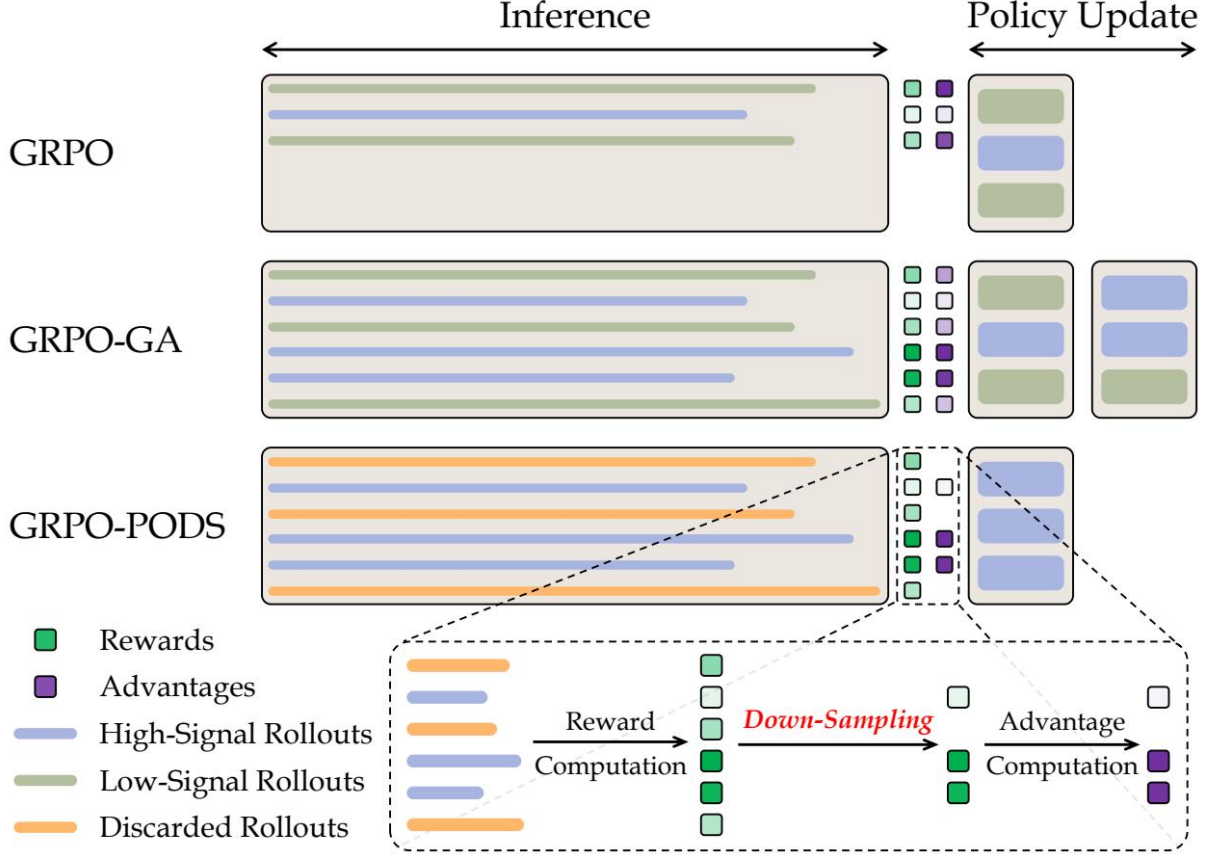


Figure 2: images/5278c9d175368e4975afd1ea1e6a455c9384e5cb9ce4a8d778919c07332f5766.jpg

Figure 2: Visualization of three training strategies: vanilla GRPO, GRPO with gradient accumulation (GRPO-GA), and GRPO with PODS (GRPO-PODS). Vanilla GRPO generates n rollouts and trains on all of them, leaving inference hardware underutilized due to the asymmetric computational demands of the two phases. GRPO-GA alleviates this issue with memory-saving techniques such as gradient accumulation, but at the cost of more sequential steps in the policy-update phase. In contrast, GRPO-PODS also generates n rollouts but trains on only m carefully selected examples, maximizing inference utilization, avoiding gradient-accumulation overhead, and providing a cleaner learning signal that yields better final performance.

图 2: 三种训练策略的可视化: 原始 GRPO、带有梯度累积的 GRPO (GRPO-GA) 以及带有 PODS 的 GRPO (GRPO-PODS)。原始 GRPO 生成 n 个 rollouts 并在所有 rollouts 上进行训练, 由于两个阶段计算需求的不对称性, 导致推理硬件利用率低下。GRPO-GA 通过诸如梯度累积等节省内存的技术缓解了这一问题, 但代价是在策略更新阶段需要更多的顺序步骤。相比之下, GRPO-PODS 也生成 n 个 rollouts, 但仅在精心选择的 m 个示例上进行训练, 从而最大化推理利用率, 避免梯度累积的开销, 并提供更清晰的学习信号, 最终性能更好。

which used large-scale RL, have sparked interest in reasoning-focused RL methods (Chen et al., 2025; Hu et al., 2025; Hu, 2025; Cui et al., 2025). Meanwhile, value-based approaches like PPO remain central (Yuan et al., 2025a,b), alongside complementary techniques such as Monte Carlo Tree Search (Gao et al., 2024; Xie et al., 2024) and multi-agent methods (FAIR et al., 2022). A recent line of work has also explored data selection for improving RL methods for LLM training. Specifically, prompt selection and filtering has gained significant attention from works such as DAPO (Yu et al., 2025), SRPO (Zhang et al., 2025), Reinforce-Rej (Xiong et al., 2025), Polaris (An et al., 2025), GRESO (Zheng et al., 2025) and VAPO (Yuan et al., 2025a). Our method advances this line of work by focusing on down-sampling rollouts within each prompt, instead of selecting or filtering prompts themselves. By tackling this computational-efficiency bottleneck, our approach

complements existing methods and can be combined with them to further improve reasoning performance.

使用大规模强化学习的方法已经引发了对以推理为中心的强化学习方法的兴趣 (Chen 等, 2025; Hu 等, 2025; Hu, 2025; Cui 等, 2025)。同时, 基于价值的方法如 PPO 仍然处于核心地位 (Yuan 等, 2025a;b), 并伴随着诸如蒙特卡洛树搜索 (Monte Carlo Tree Search) (Gao 等, 2024; Xie 等, 2024) 和多智能体方法 (FAIR 等, 2022) 等互补技术。最近有一系列工作也探索了用于改进用于大语言模型 (LLM) 训练的强化学习方法的数据选择。具体而言, 提示选择和过滤已经引起了诸如 DAPO (Yu 等, 2025)、SRPO (Zhang 等, 2025)、Reinforce-Rej (Xiong 等, 2025)、Polaris (An 等, 2025)、GRESO (Zheng 等, 2025) 和 VAPO (Yuan 等, 2025a) 等工作的广泛关注。我们的方法通过专注于在每个提示内对 rollout 进行下采样, 而不是选择或过滤提示本身, 从而推进了这一研究方向。通过解决这一计算效率瓶颈, 我们的方法可以与现有方法相结合, 进一步提升推理性能。

Down-sampling and data selection. The scale of modern machine learning necessitates effective data management strategies, particularly as datasets grow larger, noisier, and more imbalanced. Training on the full dataset can be prohibitively expensive, motivating sophisticated data-selection and down-sampling methods. Such techniques succeed across diverse settings—from theoretical results in clustering (Har-Peled & Mazumdar, 2004), regression (Li et al., 2013; Rudelson & Vershynin, 2007; Clarkson, 2010) to practical systems in speech recognition (Liu et al., 2015; Wei et al., 2014) and computer vision (Kaushal et al., 2019;

下采样和数据选择。现代机器学习的规模需要有效的数据管理策略, 尤其是在数据集变得更大、更嘈杂和更不平衡的情况下。在完整数据集上进行训练可能会耗费巨大, 这促使了复杂的数据选择和下采样方法。这些技术在各种环境中都取得了成功——从聚类 (Har-Peled & Mazumdar, 2004) 的理论结果, 回归 (Li et al., 2013; Rudelson & Vershynin, 2007; Clarkson, 2010) 到语音识别 (Liu et al., 2015; Wei et al., 2014) 和计算机视觉 (Kaushal et al., 2019;

Bankes et al., 2024). In reinforcement learning, prioritized experience replay (Schaul et al., 2015) and related methods (Hou et al., 2017; Saglam et al., 2023; Cusumano-Towner et al., 2025) highlight the value of selective sampling from experience buffers. More recently, careful data selection has become central to foundation-model training (Goyal et al., 2024; Schuhmann et al., 2021; Gadre et al., 2023) and emerging applications such as computational advertising (Bei et al., 2023; Gravin et al., 2024). Yet, to our knowledge, we are the first to apply principled down-sampling to the rollout-generation stage of LLM reinforcement learning, mitigating a key computational bottleneck while strengthening the learning signal.

Bankes 等人, 2024)。在强化学习中, 优先经验回放 (Schaul 等人, 2015) 及相关方法 (Hou 等人, 2017; Saglam 等人, 2023; Cusumano-Towner 等人, 2025) 突出了从经验缓冲区中进行选择性采样的价值。最近, 仔细的数据选择已成为基础模型训练 (Goyal 等人, 2024; Schuhmann 等人, 2021; Gadre 等人, 2023) 以及诸如计算广告 (Bei 等人, 2023; Gravin 等人, 2024) 等新兴应用的核心。然而, 据我们所知, 我们是首个将原理性的下采样应用于大语言模型强化学习的 rollout 生成阶段的研究, 这在缓解关键计算瓶颈的同时, 也加强了学习信号。

3 Down-Sampling Rollouts in GRPO

3 下采样轨迹在 GRPO 中

In this section, we present our approach to resolving the computational asymmetry between inference and policy updates in LLM reinforcement learning. We first review the original GRPO algorithm in Section 3.1, highlighting its structural components and computational demands. Next, in Section 3.2, we introduce the PODS (Policy Optimization with Down-Sampling) framework, which strategically selects informative rollouts to maximize hardware utilization during both inference and policy-update phases. In Section 3.3, we develop a principled max-variance down-sampling method that preserves strong contrastive signals, justified by Razin et al. (2025), by retaining only rollouts from the extremes of the reward spectrum. We show that this method admits an elegant, $O(n \log n)$ solution, making it practical for real-world deployment.

在本节中，我们介绍了解决大型语言模型强化学习中推理与策略更新之间计算不对称性的方法。我们首先在第 3.1 节回顾原始的 GRPO 算法，突出其结构组件和计算需求。接下来，在第 3.2 节中，我们介绍 PODS（基于下采样的策略优化）框架，该框架通过战略性地选择具有信息量的轨迹，在推理和策略更新阶段最大化硬件利用率。在第 3.3 节中，我们开发了一种基于最大方差下采样的方法，通过保留奖励谱极值处的轨迹来保持强对比信号，这一方法得到了 Razin 等人（2025）的证明。我们展示了该方法具有优雅的 $O(n \log n)$ 解决方案，使其适用于实际部署。

3.1 Preliminaries

3.1 基础知识

Group Relative Policy Optimization (GRPO) (Shao et al., 2024) is a reinforcement-learning algorithm intended to enhance the reasoning capabilities of large language models (LLMs), particularly within the RLVR setting. Each GRPO training step follows a structured, two-phase process, described below.

Group Relative Policy Optimization (GRPO) (Shao et al., 2024) 是一种强化学习算法，旨在提升大语言模型（LLMs）的推理能力，特别是在 RLVR 设置中。每个 GRPO 训练步骤遵循一个结构化的、两阶段的过程，如下所述。

Inference phase. Let π_θ denote the policy parameterized by θ , which defines a distribution over next-token probabilities given the previous tokens in a sequence. Given a single input prompt p (e.g., a math problem), GRPO first generates a group of n rollouts $\mathbf{o} = (o_1, o_2, \dots, o_n)$ by autoregressively sampling from π_θ . Each rollout is a complete token sequence excluding the prompt, representing a possible solution. Each rollout is then evaluated using a reward model $r_i = R(o_i)$, which scores the quality and correctness of the corresponding output o_i . This yields a reward vector $\mathbf{r} = (r_1, r_2, \dots, r_n)$. We then compute normalized advantage estimates: $a_i = (r_i - \mu)/\sigma$, where μ and σ are the mean and standard deviation of the rewards respectively.

推理阶段。 令 π_θ 表示由 θ 参数化的策略，它定义了一个分布，该分布根据序列中先前的标记来确定下一个标记的概率。给定一个单一的输入提示 p （例如，一个数学问题），GRPO 首先通过自回归地从 π_θ 中进行采样，生成一组 n 个 rollout $\mathbf{o} = (o_1, o_2, \dots, o_n)$ 。每个 rollout 是一个完整的标记序列，不包括提示，表示一个可能的解决方案。然后，每个 rollout 使用奖励模型 $r_i = R(o_i)$ 进行评估，该模型对相应的输出 o_i 进行评分，评估其质量和正确性。这产生了一个奖励向量 $\mathbf{r} = (r_1, r_2, \dots, r_n)$ 。接着，我们计算归一化的优势估计值： $a_i = (r_i - \mu)/\sigma$ ，其中 μ 和 σ 分别是奖励的均值和标准差。

Policy update phase. After computing the advantages, the policy is updated by optimizing the GRPO objective $L_{\text{GRPO}}(\theta)$. Specifically, for each rollout o_i with advantage a_i , we compute a loss for each token position t , and then average over all tokens and rollouts:

策略更新阶段。 在计算出优势值后，通过优化 GRPO 目标 $L_{\text{GRPO}}(\theta)$ 来更新策略。具体来说，对于每个 rollout o_i ，其优势值为 a_i ，我们计算每个 token 位置 t 的损失，然后对所有 token 和 rollout 进行平均：

$$L_{\text{GRPO}}(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left[\frac{\pi_\theta(o_{i,t}|p, o_{i,<t})}{\pi_{\theta_{\text{fixed}}}(o_{i,t}|p, o_{i,<t})} \cdot a_i, \text{clip} \left(\frac{\pi_\theta(o_{i,t}|p, o_{i,<t})}{\pi_{\theta_{\text{fixed}}}(o_{i,t}|p, o_{i,<t})}, 1 - \epsilon, 1 + \epsilon \right) \cdot a_i \right].$$

where $|o_i|$ is the number of tokens in o_i and $\pi_{\theta_{\text{fixed}}}$ is a frozen copy of the policy used for importance weighting. This asymmetric loss embodies the slow to adopt, quick to abandon learning principle—limiting how aggressively the policy increases probabilities for tokens in high-reward rollouts while allowing more substantial reductions for low-reward sequences.

其中 $|o_i|$ 是 o_i 中的 token 数量， $\pi_{\theta_{\text{fixed}}}$ 是用于重要性加权的策略的冻结副本。这种不对称损失体现了“缓慢采纳、快速放弃”的学习原则——限制策略在高奖励 rollout 中对 token 提高概率的激进程度，同时

允许在低奖励序列中进行更显著的概率降低。

3.2 PODS Framework

3.2 PODS 框架

We propose to decouple the inference and training phases in GRPO. Rather than updating on every generated rollout, PODS first produces n rollouts in parallel and then trains on only a smaller subset of size $m < n$ selected by a down-sampling rule D . This strategy exploits parallelism during inference while substantially reducing the communication and memory costs of the subsequent policy update.

我们建议在 GRPO 中将推理和训练阶段解耦。与在每次生成 rollout 后都进行更新不同，PODS 首先并行地生成 n 个 rollout，然后仅使用由下采样规则 D 选出的大小为 $m < n$ 的较小子集进行训练。这种策略在推理过程中利用了并行性，同时显著降低了后续策略更新的通信和内存成本。

Definition 3.1 (Down-sampling rule). $D(\mathbf{o}, \mathbf{r}; m)$ is a function that takes n rollouts $\mathbf{o} = (o_1, o_2, \dots, o_n)$, their corresponding rewards $\mathbf{r} = (r_1, r_2, \dots, r_n)$, and the update size m . It outputs a subset of indices $S \subseteq \{1, 2, \dots, n\}$, where $|S| = m$, indicating which rollouts to retain for the policy update phase.

定义 3.1 (下采样规则)。 $D(\mathbf{o}, \mathbf{r}; m)$ 是一个函数，它接受 n 条轨迹 $\mathbf{o} = (o_1, o_2, \dots, o_n)$ ，它们对应的奖励 $\mathbf{r} = (r_1, r_2, \dots, r_n)$ ，以及更新规模 m 。它输出一个索引子集 $S \subseteq \{1, 2, \dots, n\}$ ，其中 $|S| = m$ ，表示哪些轨迹将被保留用于策略更新阶段。

Given a selected subset of indices S , we compute the advantage estimates using only the selected rollouts: $a_{S,i} = (r_i - \mu_S) / \sigma_S$, where μ_S and σ_S are the mean and standard deviation of the rewards in the selected subset. The GRPO-PODS objective then becomes:

给定一个选定的索引子集 S ，我们仅使用选定的 rollout 来计算优势估计值： $a_{S,i} = (r_i - \mu_S) / \sigma_S$ ，其中 μ_S 和 σ_S 是选定子集中奖励的均值和标准差。GRPO-PODS 目标函数则变为：

$$L_{\text{PODS}}(\theta, S) = \frac{1}{m} \sum_{i \in S} \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left[\frac{\pi_{\theta}(o_{i,t}|p, o_{i,<t})}{\pi_{\theta_{\text{fixed}}}(o_{i,t}|p, o_{i,<t})} \cdot a_{S,i}, \text{clip} \left(\frac{\pi_{\theta}(o_{i,t}|p, o_{i,<t})}{\pi_{\theta_{\text{fixed}}}(o_{i,t}|p, o_{i,<t})}, 1 - \varepsilon, 1 + \varepsilon \right) \cdot a_{S,i} \right].$$

Algorithm 1 The PODS Framework for GRPO

算法 1 GRPO 的 PODS 框架

Input: Models $\pi_{\theta}, \pi_{\theta_{\text{fixed}}}$, input prompt p , reward model R , Number of rollouts n , update size m , down-sampling rule D Independently sample n rollouts $\mathbf{o} = (o_1, o_2, \dots, o_n)$ using $\pi_{\theta_{\text{fixed}}}$ Compute rewards $\mathbf{r} = (r_1, r_2, \dots, r_n)$ using the reward model R Down-sample a set of m rollouts $S \leftarrow D(\mathbf{o}, \mathbf{r}; m)$ Update the policy using the GRPO-PODS objective $L_{\text{PODS}}(\theta, S)$

输入：模型 $\pi_{\theta}, \pi_{\theta_{\text{fixed}}}$ ，输入提示 p ，奖励模型 R ，滚出次数 n ，更新规模 m ，下采样规则 D 独立地使用 $\pi_{\theta_{\text{fixed}}}$ 采样 n 滚出 $\mathbf{o} = (o_1, o_2, \dots, o_n)$ 使用奖励模型 R 计算奖励 $\mathbf{r} = (r_1, r_2, \dots, r_n)$ 对一组 m 滚出 $S \leftarrow D(\mathbf{o}, \mathbf{r}; m)$ 进行下采样使用 GRPO-PODS 目标更新策略 $L_{\text{PODS}}(\theta, S)$

Output: An updated model $\pi_{\theta_{\text{updated}}}$

输出：更新的模型 $\pi_{\theta_{\text{updated}}}$

Algorithm 1 outlines the PODS framework for GRPO with a single prompt p in a training iteration. When training on a batch of multiple prompts, we simply apply the same procedure to each prompt and then concatenate the down-sampled rollouts and rewards. We conclude this section by presenting two trivial down-sampling strategies that can potentially be applied within PODS.

算法 1 概述了在单个提示 p 的训练迭代中，GRPO 的 PODS 框架。当在多个提示的批次上进行训练时，我们只需对每个提示应用相同的步骤，然后连接下采样的轨迹和奖励。我们通过介绍两种简单的下采样策略来结束本节，这些策略可能在 PODS 中应用。

Random down-sampling. The rule D_{rand} uniformly selects m indices from $\{1, 2, \dots, n\}$ without replacement, thereby preserving the statistical properties of the original rollout distribution. In expectation, it yields the same parameter update as running standard GRPO on exactly m rollouts.

随机下采样。规则 D_{rand} 均匀地从 $\{1, 2, \dots, n\}$ 中选择 m 个不重复的索引，从而保留原始 rollout 分布的统计特性。在期望上，它产生的参数更新与在恰好 m 个 rollout 上运行标准 GRPO 是相同的。

Percentile down-sampling. The rule D_{perc} selects the m rollouts by choosing the $\frac{0.5}{m}, \frac{1.5}{m}, \dots, \frac{m-0.5}{m}$ quantiles of the reward distribution. This method ensures that the selected rollouts are evenly spaced across the reward spectrum, providing a more representative sample than random down-sampling.

百分位数下采样。规则 D_{perc} 通过选择奖励分布的 $\frac{0.5}{m}, \frac{1.5}{m}, \dots, \frac{m-0.5}{m}$ 分位数来选择 m 个 rollout。这种方法确保所选的 rollout 在奖励范围内均匀分布，比随机下采样提供更具代表性的样本。

Max-reward down-sampling. The rule D_{maxr} selects the m rollouts with the highest rewards, concentrating on examples that exhibit the most desirable behavior. This should allow the model to learn primarily from successful reasoning patterns. However, as we show in Section 4, ignoring low-reward rollouts deprives the policy of negative feedback and can significantly degrade performance.

最大奖励下采样。规则 D_{maxr} 选择具有最高奖励的 m 个轨迹，专注于表现出最理想行为的例子。这应该使模型主要从成功的推理模式中学习。然而，如我们在第 4 节中所展示的，忽略低奖励轨迹会使策略失去负反馈，并可能显著降低性能。

3.3 Max-Variance Down-Sampling

3.3 最大方差下采样

We now introduce max-variance down-sampling, a principled down-sampling rule that selects the most diverse and informative rollouts according to their reward distribution.

我们现在介绍最大方差下采样（max-variance down-sampling），这是一种有原则的下采样规则，根据奖励分布选择最具多样性和信息量的 rollouts。

Specifically, D_{maxv} chooses the subset S of size m that maximizes the empirical reward variance, i.e., $S = \arg \max_{|S|=m} \text{Var}(\{r_i \mid i \in S\})$. By spanning the full performance spectrum, it supplies strong contrastive signals between successful and unsuccessful reasoning paths. Recent work by Razin et al. (2025) provides an optimization-theoretic and empirical justification for this criterion.

具体而言， D_{maxv} 选择大小为 m 的子集 S ，以最大化经验奖励方差，即 $S = \arg \max_{|S|=m} \text{Var}(\{r_i \mid i \in S\})$ 。通过覆盖完整的性能范围，它在成功与不成功的推理路径之间提供了强对比信号。Razin 等人 (2025) 的近期研究从优化理论和实证角度为这一标准提供了依据。

A naive search would examine $O(\binom{n}{m})$ subsets. This is clearly infeasible for realistic n and m . We prove, however, that the optimal subset can be found in $O(n \log n)$ time.

一种简单的搜索方法会检查 $O(\binom{n}{m})$ 个子集。这对于现实中的 n 和 m 显然是不可行的。然而，我们证明了最优子集可以在 $O(n \log n)$ 时间内找到。

Lemma 3.1. For a sorted list of rewards $r_1 \leq r_2 \leq \dots \leq r_n$, the variance-maximizing subset of size m always consists of the k highest rewards and $(m - k)$ lowest rewards for some $k \in \{0, 1, \dots, m\}$. That is,

引理 3.1. 对于一个排序后的奖励列表 $r_1 \leq r_2 \leq \dots \leq r_n$ ，大小为 m 的方差最大化子集总是由 k 个最高奖励和 $(m - k)$ 个最低奖励组成，其中 $k \in \{0, 1, \dots, m\}$ 。也就是说，

$$\text{Var}(\{r_1, \dots, r_{m-k}\} \cup \{r_{n-k+1}, \dots, r_n\}) = \max_{|S|=m} \text{Var}(\{r_i \mid i \in S\}).$$

Proof of Lemma 3.1: Let $S^* = \arg \max_{|S|=m} \text{Var}(\{r_i \mid i \in S\})$ be the optimal subset of size m . We will show that if S^* is not of the form $\{1, \dots, m - k\} \cup \{n - k + 1, \dots, n\}$ for any k , then we can modify S^* to obtain a new subset S' of the same size with no smaller variance in rewards. By repeating this procedure, we can eventually reach a subset of this form.

引理 3.1 的证明: 设 $S^* = \arg \max_{|S|=m} \text{Var}(\{r_i \mid i \in S\})$ 是大小为 m 的最优子集。我们将证明，如果 S^* 不是任何 k 的形式 $\{1, \dots, m - k\} \cup \{n - k + 1, \dots, n\}$ ，那么我们可以修改 S^* 以得到一个大小相同的新子集 S' ，其奖励方差不大于原子集。通过重复这一过程，我们最终可以得到一个具有这种形式的子集。

Let μ be the mean of the rewards in S^* . Since S^* does not take the form of $\{1, \dots, m - k\} \cup \{n - k + 1, \dots, n\}$ for any k , there exists either (i) an element $i \in S^*$ such that $i > 1, r_i \leq \mu$ and $i - 1 \notin S^*$, or (ii) an element $j \in S^*$ such that $j < n, r_j \geq \mu$ and $j + 1 \notin S^*$. That is, there exists an element in S^* , such that another element further from μ is not in S^* . We will show that we can swap them without decreasing variance.

设 μ 为 S^* 中的奖励均值。由于对于任何 k ， S^* 并不具有 $\{1, \dots, m - k\} \cup \{n - k + 1, \dots, n\}$ 的形式，因此存在以下两种情况之一：(i) 一个元素 $i \in S^*$ ，使得 $i > 1, r_i \leq \mu$ 且 $i - 1 \notin S^*$ ，或 (ii) 一个元素 $j \in S^*$ ，使得 $j < n, r_j \geq \mu$ 且 $j + 1 \notin S^*$ 。也就是说，存在一个元素在 S^* 中，使得另一个距离 μ 更远的元素不在 S^* 中。我们将证明，我们可以交换它们而不减少方差。

For the ease of notation, we will denote $\text{Var}(\{r_i \mid i \in S\})$ as $\text{Var}(S)$ in this proof.

为了便于表示，在这个证明中，我们将 $\text{Var}(\{r_i \mid i \in S\})$ 表示为 $\text{Var}(S)$ 。

For case (i), let $S' = (S^* \setminus \{i\}) \cup \{i - 1\}$, and let μ' be the mean of the rewards in S' . Then

对于情况 (i)，设 $S' = (S^* \setminus \{i\}) \cup \{i - 1\}$ ，并设 μ' 为 S' 中的奖励的平均值。然后

$$\begin{aligned} \text{Var}(S') - \text{Var}(S^*) &= \left(\frac{1}{m} \sum_{t \in S'} r_t^2 - \mu'^2 \right) - \left(\frac{1}{m} \sum_{t \in S^*} r_t^2 - \mu^2 \right) \\ &= \frac{1}{m} (r_{i-1}^2 - r_i^2) - (\mu'^2 - \mu^2) \\ &= \frac{1}{m} (r_{i-1} - r_i)(r_{i-1} + r_i) - (\mu' - \mu)(\mu' + \mu) \\ &= \frac{1}{m} (r_{i-1} - r_i)[(r_{i-1} + r_i) - (\mu' + \mu)] \geq 0. \end{aligned}$$

For case (ii), let $S' = (S^* \setminus \{j\}) \cup \{j + 1\}$, we can similarly show that $\text{Var}(S') - \text{Var}(S^*) \geq 0$.

对于情况 (ii)，令 $S' = (S^* \setminus \{j\}) \cup \{j + 1\}$ ，我们可以类似地证明 $\text{Var}(S') - \text{Var}(S^*) \geq 0$ 。

In either case, we have shown that we can modify S^* to obtain a new subset S' of the same size that has no smaller variance in rewards. We can repeat this process until we reach a subset of the form $\{1, \dots, m - k\} \cup \{n - k + 1, \dots, n\}$ for some k . Thus, we conclude that there must exist one optimal subset of this form for some k .

在任何情况下, 我们已经证明, 我们可以修改 S^* , 以获得一个大小相同的新的子集 S' , 其奖励的方差不小于原子集。我们可以重复这一过程, 直到得到一个形如 $\{1, \dots, m - k\} \cup \{n - k + 1, \dots, n\}$ 的子集, 其中 k 是某个值。因此, 我们得出结论, 必须存在一个这样的最优子集, 其形式为 k 。

Lemma 3.1 naturally leads to a practical algorithm, Algorithm 2, for max-variance down-sampling. Moreover, it also offers intuition as to why maximizing variance is effective: the optimal subset contains the k highest and the $(m - k)$ lowest rewards, capturing contrastive signals from both positive and negative examples.

引理 3.1 自然地引导出一个实用的算法, 即算法 2, 用于最大方差下采样。此外, 它还提供了为什么最大化方差是有效的直觉: 最优子集包含 k 个最高的奖励和 $(m - k)$ 个最低的奖励, 从而从正例和负例中捕捉到对比信号。

Algorithm 2 Max-Variance Down-Sampling

算法 2 最大方差下采样

Input: Number of rollouts n , update size m , rollouts $\{o_1, o_2, \dots, o_n\}$, rewards $\{r_1, r_2, \dots, r_n\}$
Sort the rollouts by reward and get the sorted indices $ind \leftarrow \text{argsort}(\{r_1, r_2, \dots, r_n\})$ Let $S_{\text{ans}} \leftarrow \{ind_1, \dots, ind_m\}$ for $k \in \{1, \dots, m\}$ do Let $S_{\text{this}} \leftarrow \{ind_1, \dots, ind_{m-k}\} \cup \{ind_{n-k+1}, \dots, ind_n\}$ Let $S_{\text{ans}} \leftarrow S_{\text{this}}$ if $\text{Var}(\{r_i \mid i \in S_{\text{this}}\}) > \text{Var}(\{r_i \mid i \in S_{\text{ans}}\})$ end for
Output: Selected indices S_{ans} of rollouts

输入: rollouts 数量 n , 更新大小 m , rollouts $\{o_1, o_2, \dots, o_n\}$, 奖励 $\{r_1, r_2, \dots, r_n\}$ 按奖励对 rollouts 进行排序, 并获取排序后的索引 $ind \leftarrow \text{argsort}(\{r_1, r_2, \dots, r_n\})$ 让 $S_{\text{ans}} \leftarrow \{ind_1, \dots, ind_m\}$ 对于 $k \in \{1, \dots, m\}$ 做让 $S_{\text{this}} \leftarrow \{ind_1, \dots, ind_{m-k}\} \cup \{ind_{n-k+1}, \dots, ind_n\}$ 让 $S_{\text{ans}} \leftarrow S_{\text{this}}$ 如果 $\text{Var}(\{r_i \mid i \in S_{\text{this}}\}) > \text{Var}(\{r_i \mid i \in S_{\text{ans}}\})$ 结束循环输出: 所选 rollouts 的索引 S_{ans}

Theorem 1. Algorithm 2 computes the max-variance down-sampling rule correctly. Moreover, it can be implemented in $O(n \log n)$ time.

定理 1. 算法 2 正确地计算了最大方差下采样规则。此外, 该算法可以在 $O(n \log n)$ 时间内实现。

Proof of Theorem 1: The correctness of Algorithm 2 follows directly from Lemma 3.1.

定理 1 的证明: 算法 2 的正确性直接由引理 3.1 得出。

For the time complexity, we first sort the rewards in $O(n \log n)$ time. To compute the variance of the selected rollouts, note that $\text{Var}(\{x \mid x \in S_{\text{this}}\}) = \mathbf{E}_{x \in S_{\text{this}}}[x^2] - (\mathbf{E}_{x \in S_{\text{this}}}[x])^2$. We can maintain the

对于时间复杂度, 我们首先在 $O(n \log n)$ 时间内对奖励进行排序。为了计算所选 rollout 的方差, 注意 $\text{Var}(\{x \mid x \in S_{\text{this}}\}) = \mathbf{E}_{x \in S_{\text{this}}}[x^2] - (\mathbf{E}_{x \in S_{\text{this}}}[x])^2$ 。我们可以维护

prefix sums of the rewards and the squared rewards in $O(n)$ time. Then, for each k , we can compute the variance of the selected rollouts in $O(1)$ time using the prefix sums. Thus, the overall time complexity is $O(n \log n) + O(m) = O(n \log n)$.

在 $O(n)$ 时间内计算奖励和平方奖励的前缀和。然后, 对于每个 k , 我们可以使用前缀和在 $O(1)$ 时间内计算所选轨迹的方差。因此, 总体时间复杂度为 $O(n \log n) + O(m) = O(n \log n)$ 。

Theorem 1 shows that the max-variance down-sampling rule can be computed efficiently, which enables its practical application in GRPO-PODS. We conclude this section by noting an important special case of the max-variance down-sampling rule.

定理 1 表明，最大方差下采样规则可以高效地进行计算，这使得其在 GRPO-PODS 中的实际应用成为可能。我们通过指出最大方差下采样规则的一个重要特殊情况来结束本节。

Theorem 2. Let m be an even integer. When the rewards are binary, selecting $m/2$ rollouts with the highest rewards and $m/2$ rollouts with the lowest rewards maximizes the variance of the rewards.

定理 2。设 m 为偶整数。当奖励为二元时，选择具有最高奖励的 $m/2$ 个 rollout 和具有最低奖励的 $m/2$ 个 rollout 可以最大化奖励的方差。

Proof of Theorem 2: Let the number of rollouts with reward 1 be k . Then, the number of rollouts with reward 0 is $n - k$. If $k \leq m/2$, then any subset of m rollouts contains at most k rollouts with reward 1, and the variance is maximized by selecting these k rollouts and any $(m - k)$ rollouts with reward 0. If $n - k \leq m/2$, then any subset of m rollouts contains at most $(n - k)$ rollouts with reward 0, and the variance is maximized by selecting these $(n - k)$ rollouts and any $m - (n - k)$ rollouts with reward 1. Otherwise, any subset of $m/2$ rollouts with reward 1 and $m/2$ rollouts with reward 0 maximizes the variance. In all cases, we can select $m/2$ rollouts with the highest rewards and $m/2$ rollouts with the lowest rewards to maximize the variance. This concludes the proof.

定理 2 的证明：设奖励为 1 的 rollouts 的数量为 k 。那么，奖励为 0 的 rollouts 的数量为 $n - k$ 。如果 $k \leq m/2$ ，那么任何包含 m 个 rollouts 的子集最多包含 k 个奖励为 1 的 rollouts，通过选择这些 k 个 rollouts 以及任意 $(m - k)$ 个奖励为 0 的 rollouts 可以最大化方差。如果 $n - k \leq m/2$ ，那么任何包含 m 个 rollouts 的子集最多包含 $(n - k)$ 个奖励为 0 的 rollouts，通过选择这些 $(n - k)$ 个 rollouts 以及任意 $m - (n - k)$ 个奖励为 1 的 rollouts 可以最大化方差。否则，任何包含 $m/2$ 个奖励为 1 的 rollouts 和 $m/2$ 个奖励为 0 的 rollouts 的子集都可以最大化方差。在所有情况下，我们都可以选择 $m/2$ 个奖励最高的 rollouts 和 $m/2$ 个奖励最低的 rollouts 以最大化方差。这完成了证明。

4 Experiments

4 实验

We evaluate PODS across diverse hardware configurations, model architectures, and model scales to demonstrate its generalizability and practical benefits. We test on three reasoning benchmarks of different domains, GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021) and the L3 Chemistry subset of SciKnowEval (Feng et al., 2024), using Qwen2.5 (Qwen et al., 2025) and Llama3.2 (MetaAI, 2024) models ranging from 3B to 7B parameters. Our experimental design covers both resource-constrained single-GPU setups and multi-GPU distributed training to validate PODS across different deployment scenarios. Table 1 describes our experimental configurations. We publish the code for our experiments at <https://github.com/YixuanEvenXu/pods>.

我们在多种硬件配置、模型架构和模型规模下评估 PODS，以展示其通用性和实际优势。我们在三个不同领域的推理基准上进行测试，包括 GSM8K (Cobbe 等, 2021)、MATH (Hendrycks 等, 2021) 以及 SciKnowEval 的 L3 Chemistry 子集 (Feng 等, 2024)，使用 Qwen2.5 (Qwen 等, 2025) 和 Llama3.2 (MetaAI, 2024) 模型，参数范围从 3B 到 7B。我们的实验设计涵盖了资源受限的单 GPU 设置和多 GPU 分布式训练，以验证 PODS 在不同部署场景下的表现。表 1 描述了我们的实验配置。我们将在 <https://github.com/YixuanEvenXu/pods> 发布我们的实验代码。

Setting	Benchmark	Model	Parameters	GPUs	Fine-tuning Method
(a)	GSM8K	Qwen2.5	3B	1 L40S	LoRA (rank 64, $\alpha = 64$)

(b)	GSM8K	Llama3.2	3B	1 L40S	LoRA (rank 64, $\alpha = 64$)
(c)	MATH	Qwen2.5	3B	1 L40S	LoRA (rank 64, $\alpha = 64$)
(d)	Chemistry	Qwen2.5	3B	1 L40S	LoRA (rank 64, $\alpha = 64$)
(e)	GSM8K	Qwen2.5	3B	8 H100s	Full-Parameter
(f)	GSM8K	Qwen2.5	7B	8 A100s	Full-Parameter

设置	基准测试	模型	参数	GPU (图形处理器)	微调方法
(a)	GSM8K	Qwen2.5	3B	1 L40S	LoRA (rank 64, $\alpha = 64$)
(b)	GSM8K	Llama3.2	3B	1 L40S	LoRA (rank 64, $\alpha = 64$)
(c)	数学	通义千问 2.5	3B	1 L40S	LoRA (rank 64, $\alpha = 64$)
(d)	化学	通义千问 2.5	3B	1 L40S	LoRA (rank 64, $\alpha = 64$)
(e)	GSM8K	通义千问 2.5	3B	8 H100s	全参数
(f)	GSM8K	通义千问 2.5	7B	8 A100s	全参数

Table 1: Experimental configurations testing GRPO with PODS across task domains, model scales, hardware constraints, and training paradigms. Settings (a-d) test resource-constrained scenarios with LoRA fine-tuning, while (e-f) evaluate full-parameter training with distributed setups.

表 1: 在不同任务领域、模型规模、硬件限制和训练范式下，测试 GRPO 与 PODS 的实验配置。设置 (a-d) 测试使用 LoRA 微调的资源受限场景，而 (e-f) 评估在分布式设置下的全参数训练。

Training infrastructure. For settings (a-d), we use Unsloth (Daniel Han & team, 2023) with TRL (von Werra et al., 2020) for efficient LoRA (Hu et al., 2022) reinforcement learning. For settings (e-f), we implement distributed training with DeepSpeed ZeRO-2 (Rajbhandari et al., 2020) and extend the open-r1 library (HuggingFace, 2025) to support PODS on multiple devices.

训练基础设施。对于设置 (a-d)，我们使用 Unsloth (Daniel Han 及其团队, 2023) 结合 TRL (von Werra 等人, 2020) 进行高效的 LoRA (Hu 等人, 2022) 强化学习。对于设置 (e-f)，我们采用 DeepSpeed ZeRO-2 (Rajbhandari 等人, 2020) 实现分布式训练，并扩展 open-r1 库 (HuggingFace, 2025) 以支持多设备上的 PODS。

Rewards and evaluations. We employ rule-based reward models that score rollouts, following standard practices in reasoning evaluation. Specifically, we reward an answer for correctness, format compliance, and the right number of thinking tags separately, resulting in a discrete but non-binary reward function. We provide additional details about the reward models Section A.1.

奖励与评估。我们采用基于规则的奖励模型来对推理过程进行评分，遵循推理评估中的标准做法。具体而言，我们分别对答案的正确性、格式合规性以及正确数量的思考标签进行奖励，从而得到一个离散但非二元的奖励函数。关于奖励模型的更多细节，请参见附录 A.1 部分。

Section roadmap. In Section 4.1, we compare the performance of GRPO and GRPO-PODS across six hardware and model settings listed in Table 1. We show that for all the settings we test, GRPO-PODS

章节路线图。在第 4.1 节中，我们比较了 GRPO 和 GRPO-PODS 在表 1 中列出的六个硬件和模型设置下的性能。我们展示了对于所有我们测试的设置，GRPO-PODS

consistently outperforms GRPO in terms of test performance throughout the training. Then, in Section 4.2, we focus on setting (a), and analyze the effect of the rollout and update sizes (n, m) on the performance of GRPO-PODS, providing empirical insights into how to choose the rollout

and update sizes for GRPO-PODS. We also present additional experiments about different down-sampling rules (Section 4.3) and different design choices of advantage normalization (Section A.3), additional evaluation results about PODS’ speed up ratio compared to GRPO (Section A.4) and test performance generalization across different test sets (Section A.5), and the average response length over the course of training (Section A.6).

在整个训练过程中，其测试性能始终优于 GRPO。然后，在第 4.2 节中，我们专注于设置 (a)，并分析 rollout 和 update 的大小 (n, m) 对 GRPO-PODS 性能的影响，提供了关于如何为 GRPO-PODS 选择 rollout 和 update 大小的经验见解。我们还展示了关于不同下采样规则（第 4.3 节）和不同优势归一化设计选择（第 A.3 节）的额外实验，以及关于 PODS 相比 GRPO 的速度提升比例（第 A.4 节）和在不同测试集上的测试性能泛化能力（第 A.5 节）的额外评估结果，以及训练过程中平均响应长度（第 A.6 节）。

4.1 Comparing GRPO-PODS to baseline GRPO

4.1 将 GRPO-PODS 与基线 GRPO 进行比较

We evaluate max-variance down-sampling PODS with GRPO against baseline GRPO using two experimental designs reflecting real-world constraints. For single-GPU settings (a-d), we compare against vanilla GRPO with the same training batch sizes (m), where m is selected so that one training batch fits within GPU memory. This corresponds to the comparison between the first and the third rows in Fig. 2. For distributed settings (e-f), we compare against GRPO with gradient accumulation (GRPO-GA), the standard approach for scaling RLVR. In GRPO-GA, large training batches are processed through multiple gradient accumulation steps, enabling updating on larger effective batch sizes at the cost of increased communication overhead and iteration time. We fix the total rollouts generated per prompt (n) and compare GRPO-GA against GRPO-PODS. This corresponds to the comparison between the second and the third rows in Fig. 2. The detailed hyperparameters used for our experiments are listed in Section A.2.

我们通过两种实验设计来评估最大方差下采样的 PODS 与 GRPO 在真实世界约束下的表现，与基线 GRPO 进行比较。对于单 GPU 设置 (a-d)，我们与使用相同训练批次大小 (m) 的普通 GRPO 进行比较，其中 m 的选择使得一个训练批次可以完全装入 GPU 内存中。这对应于图 2 中第一行与第三行的比较。对于分布式设置 (e-f)，我们与使用梯度累积的 GRPO (GRPO-GA) 进行比较，这是扩展强化学习 (RLVR) 的标准方法。在 GRPO-GA 中，通过多个梯度累积步骤处理大型训练批次，使得可以在增加通信开销和迭代时间的情况下使用更大的有效批次大小进行更新。我们固定每个提示生成的总轨迹数 (n)，并比较 GRPO-GA 与 GRPO-PODS。这对应于图 2 中第二行与第三行的比较。我们实验中使用的详细超参数列在第 A.2 节中。

Fig. 3 shows test accuracy over wall-clock training time across all experiment configurations. PODS consistently achieves faster convergence: reaching the baselines’ peak accuracies at least $1.7\times$ faster (see Section A.4 for complete results) while often converging to a higher final performance. These results demonstrate PODS’ broad applicability across task domains (Math and Chemistry), model scales (3B-7B), architectures (Qwen2.5, Llama3.2), and deployment scenarios, making it a practical improvement for RLVR systems using GRPO.

图 3 展示了所有实验配置下，测试准确率随实际训练时间的变化情况。PODS 始终实现更快的收敛速度：在至少 $1.7\times$ 的时间内达到基线的峰值准确率（详见第 A.4 节以获取完整结果），并且通常收敛到更高的最终性能。这些结果表明，PODS 在任务领域（数学和化学）、模型规模（3B-7B）、架构（Qwen2.5、Llama3.2）以及部署场景方面具有广泛适用性，使其成为使用 GRPO 的 RLVR 系统的实用改进。

4.2 Effect of Rollout and Update Sizes (n, m)

4.2 滚出和更新规模的影响 (n, m)

A key practical question for PODS adoption is how to choose the rollout size (n) and training batch size (m). While a larger n provides more diverse rollouts for selection, it also increases inference costs. Meanwhile, a smaller m reduces update costs but may provide insufficient training signal.

As shown in Fig. 4, we study these trade-offs on experimental setting (a) to provide deployment guidance.

PODS 采用中的一个关键实践问题是如何选择 rollout 的规模 (n) 和训练批次大小 (m)。虽然较大的 n 可以为选择提供更多的多样化 rollout，但它也会增加推理成本。同时，较小的 m 虽然可以降低更新成本，但可能会提供不足的训练信号。如图 4 所示，我们在实验设置 (a) 中研究了这些权衡，以提供部署指导。

Increasing rollout size n exhibits diminishing returns with an optimal point around $n = 64$. Performance initially improves as larger pools enable better sample selection, but degrades beyond $n = 128$ due to two factors: (1) inference runtime grows significantly as GPU memory saturates, and (2) marginal improvements in rollout diversity plateau while computational overhead continues rising.

增加 rollout 的规模 n 会带来递减的回报，其最佳点大约在 $n = 64$ 附近。性能最初随着更大的池 (pool) 而改善，因为更大的池能够实现更好的样本选择；但当 n 超过 128 时，性能会下降，这是由于两个因素：(1) 随着 GPU 内存达到饱和，推理运行时间显著增加；(2) rollout 多样性的边际改进趋于平稳，而计算开销却持续上升。

Training batch size m shows robust performance across a wide range, with minimal degradation until very small values ($m \leq 4$). This suggests PODS' max-variance selection maintains effective learning signals even with aggressive down-sampling ratios up to 16 where $n = 64, m = 4$.

训练批次大小 m 在一个广泛的范围内表现出鲁棒性，直到非常小的值 ($m \leq 4$) 时降级也很小。这表明 PODS 的最大方差选择即使在 $n = 64, m = 4$ 的情况下，使用高达 16 的激进下采样比率时仍能保持有效的学习信号。

Practical guidelines. These results suggest down-sampling ratio of 2 to 4 provides an effective balance of performance and efficiency. For resource-constrained settings, aggressive down-sampling ratios up to 16 remain viable, while memory-rich environments can benefit from larger rollout pools up to hardware limits.

实用指南。这些结果表明，下采样比例在 2 到 4 之间可以提供性能和效率之间的有效平衡。在资源受限的环境中，激进的下采样比例甚至可以达到 16，仍然可行；而在内存丰富的环境中，可以利用更大的 rollout 池，直到达到硬件限制。

4.3 Comparing Different Down-Sampling Rules

4.3 比较不同的下采样规则

We study the effect of different down-sampling rules on the performance of GRPO-PODS in this section. We conduct experiments on experimental setting (a). We compare four different down-sampling rules: (1) max-variance down-sampling, (2) max-reward down-sampling, (3) random down-sampling, and (4) percentile down-sampling. The results are shown in Fig. 5. We observe that the max-variance down-sampling rule consistently outperforms the other down-sampling rules throughout the training. This indicates that the max-variance down-sampling rule is effective in selecting informative rollouts for training.

我们在本节研究不同的下采样规则对 GRPO-PODS 性能的影响。我们在实验设置 (a) 上进行实验。我们比较了四种不同的下采样规则：(1) 最大方差下采样，(2) 最大奖励下采样，(3) 随机下采样，以及 (4) 百分位数下采样。结果如图 5 所示。我们观察到，最大方差下采样规则在整个训练过程中始终优于其他下采样规则。这表明最大方差下采样规则在选择用于训练的信息性 rollout 方面是有效的。

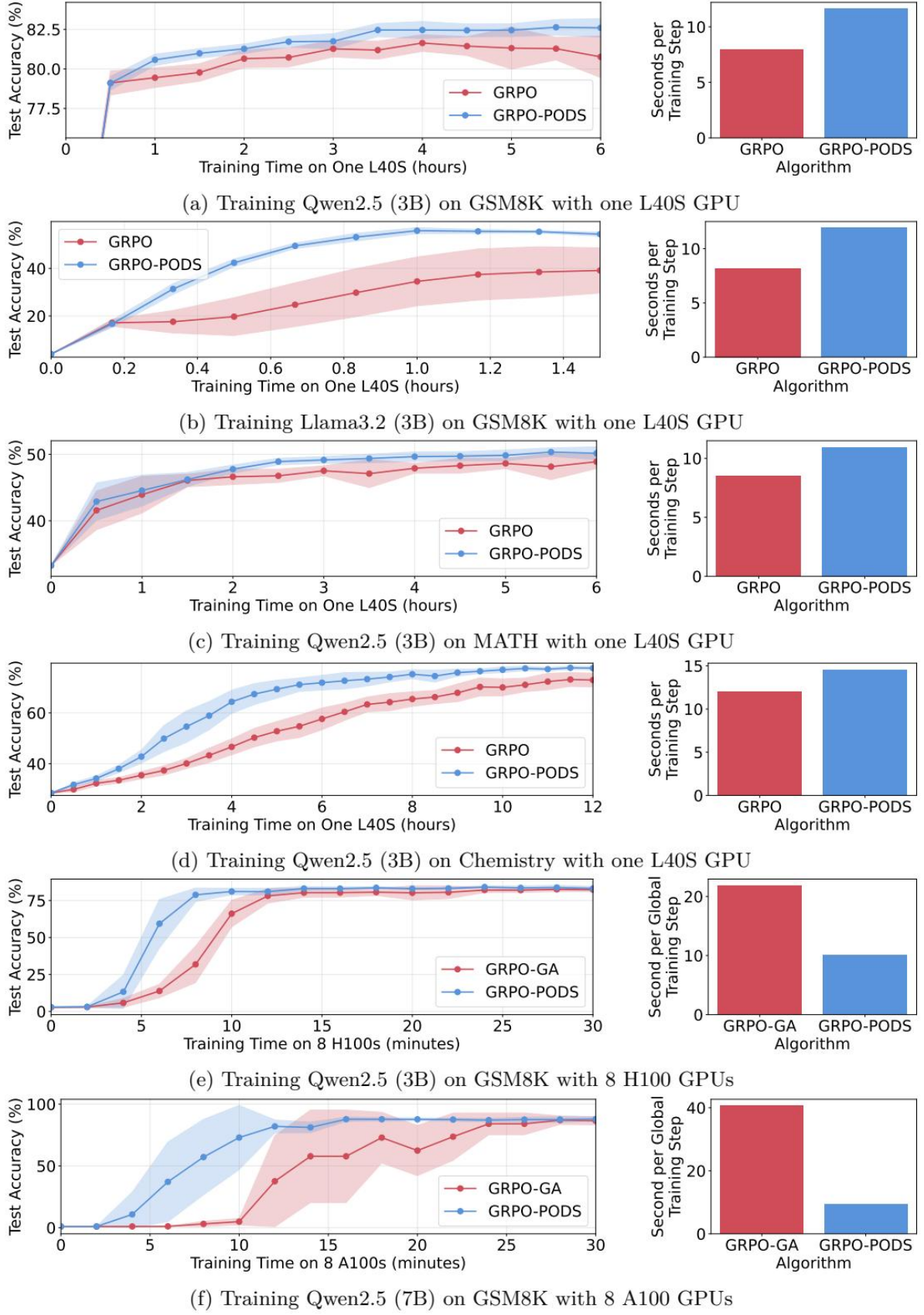


Figure 3: images/8154a45c61ce7f55ffeb2125f740998a0de00659c4f1a52ee3fc62136267aa4f.jpg

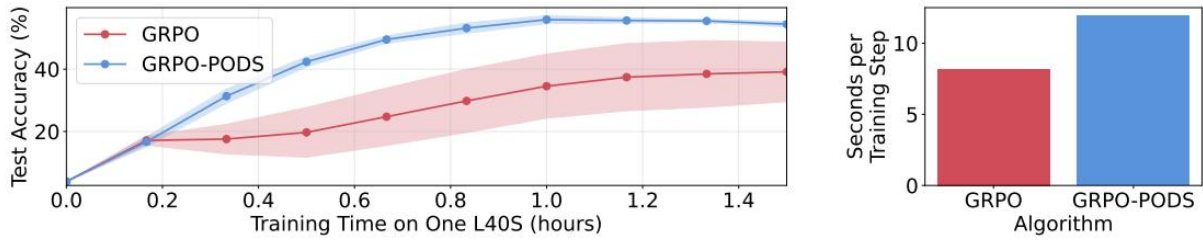


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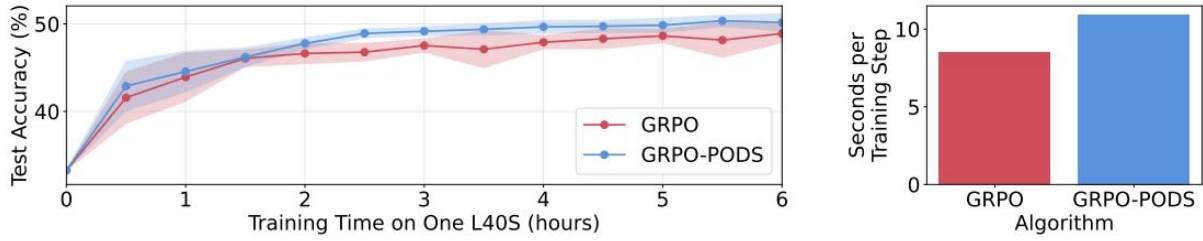


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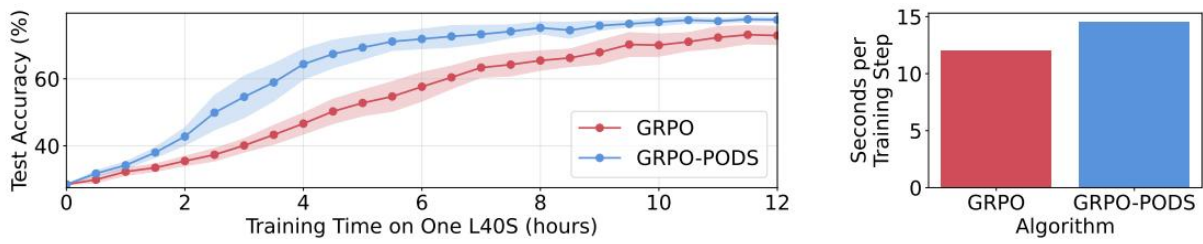


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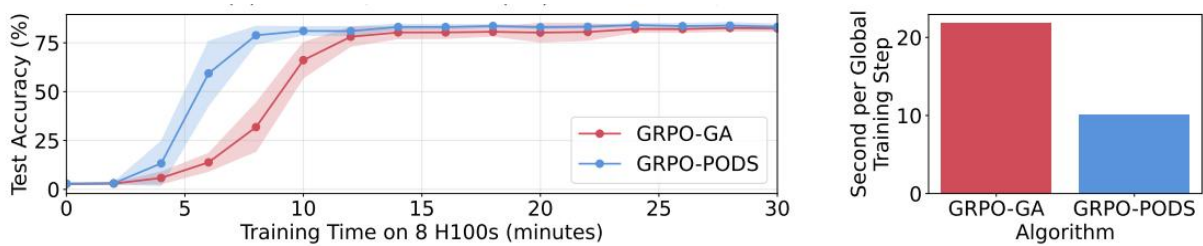


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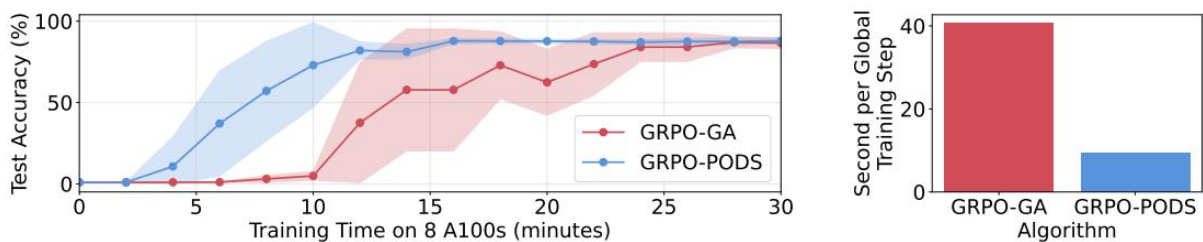


Figure 3: Performance and per-step run time comparison of standard GRPO and GRPO-PODS with

max-variance down-sampling across different datasets and hardware environments. For the performance comparison, the x-axis shows the training time, and the y-axis shows the accuracy on the test set. The shaded area represents 1.96 times the standard error of the mean.

图 3: 标准 GRPO 和 GRPO-PODS 在不同数据集和硬件环境下的性能及每步运行时间比较 (使用最大方差下采样)。在性能比较中, x 轴表示训练时间, y 轴表示测试集上的准确率。阴影区域表示均值的标准误差的 1.96 倍。

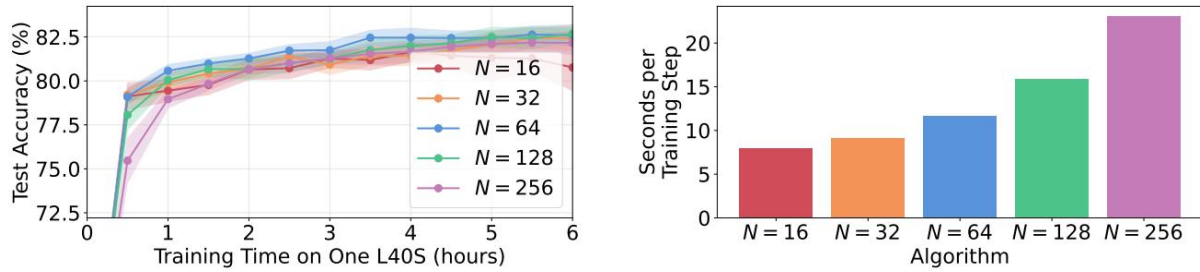


Figure 8: images/14ac44bd9467dcc8115e194a17191ac4f64dcde00e8dcbf185d883cc280658f5.jpg

(a) Fixing $m = 16$ and varying $n \in \{16, 32, 64, 128, 256\}$

(a) 修复 $m = 16$ 并改变 $n \in \{16, 32, 64, 128, 256\}$

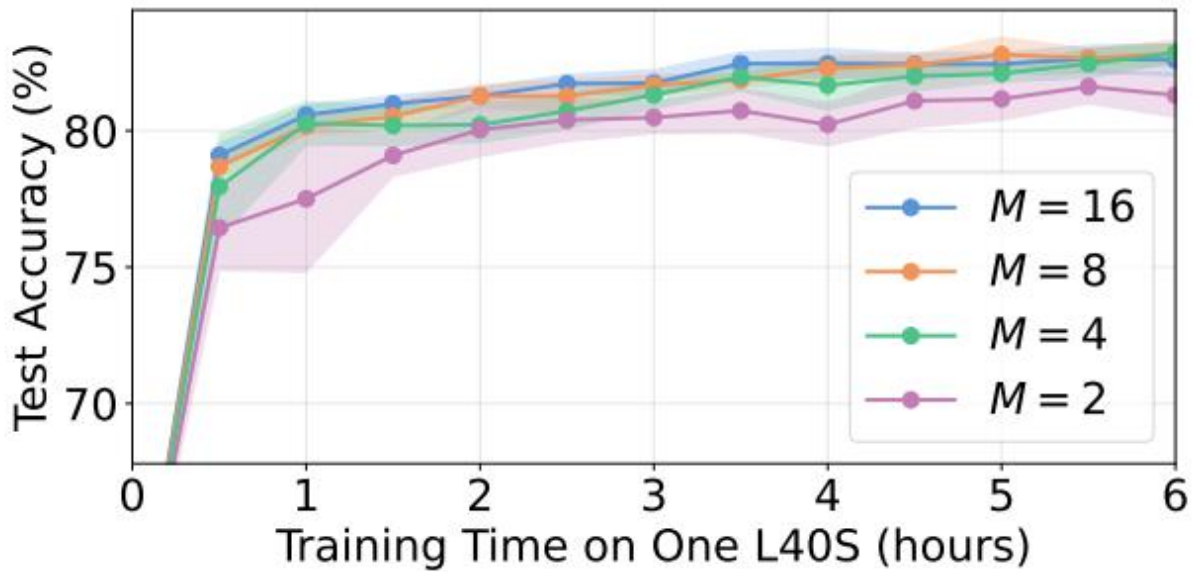


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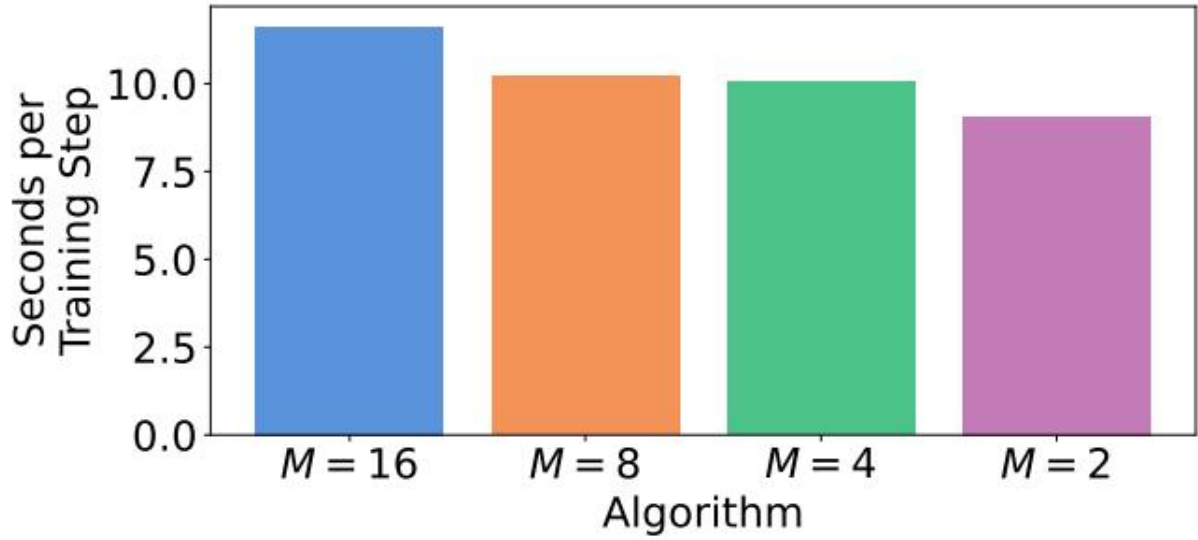


Figure 10: images/94c9ba6d9095cc5f87a45a939ba82cd0eebb5705925b044499c7291dda0797e6.jpg

(b) Fixing $n = 64$ and varying $m \in \{16, 8, 4, 2\}$

(b) 修复 $n = 64$ 并改变 $m \in \{16, 8, 4, 2\}$

Figure 4: Performance and per-step run time comparison of GRPO-PODS with max-variance down-sampling across different settings of n and m . The training is conducted on the GSM8K dataset with one L40S. For the performance comparison, the x-axis shows the training time, and the y-axis shows the accuracy on the test set. The shaded area represents 1.96 times the standard error of the mean.

图 4: GRPO-PODS 在不同 n 和 m 设置下使用最大方差下采样的性能和每步运行时间比较。训练是在配备一个 L40S 的 GSM8K 数据集上进行的。在性能比较中, x 轴表示训练时间, y 轴表示测试集上的准确率。阴影区域表示均值的标准误差的 1.96 倍。

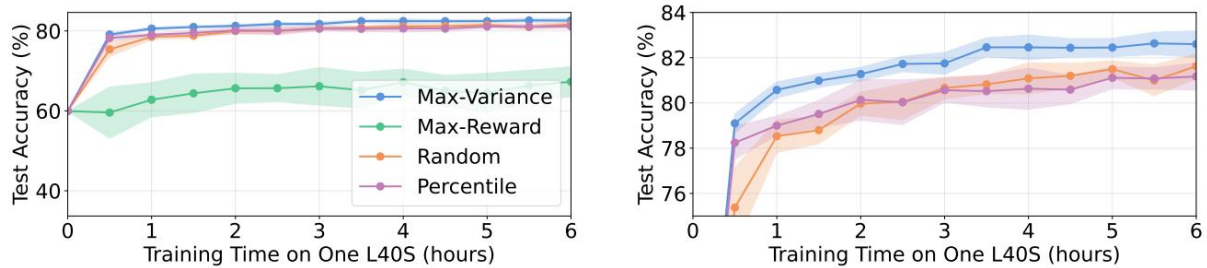


Figure 5: Performance of GRPO-PODS with the max-variance, max-reward, random and percentile down-sampling rules. The training is conducted on the GSM8K dataset with one L40S. The x-axis shows the training time, and the y-axis shows the accuracy on the test set. The shaded area represents 1.96 times the standard error of the mean. The right panel shows a Zoomed-in view of the left one.

图 5: 使用最大方差、最大奖励、随机和百分位数下采样规则的 GRPO-PODS 性能。训练是在配备一个 L40S 的 GSM8K 数据集上进行的。x 轴表示训练时间, y 轴表示测试集上的准确率。阴影区域代表均值的标准误差的 1.96 倍。右侧面板展示了左侧面板的放大视图。

5 Conclusion and Discussion

5 结论与讨论

We introduced PODS, a lightweight framework that addresses a fundamental bottleneck in modern RLVR training: the asymmetry between embarrassingly parallel rollout generation and memory-intensive policy updates. PODS generates large batches of rollouts in parallel and updates the policy on only an informative subset chosen by the max-variance rule. Our analysis shows that the optimal subset can be found in $O(n \log n)$ time. This simple yet principled approach consistently outperforms standard GRPO under equal wall-clock budgets, delivers at least a $1.7\times$ speedup and reaching higher final accuracy across diverse model architectures, scales, and deployment scenarios. Our ablation study shows that the performance of PODS is robust over a wide range of down-sampling ratios provided m is not too small, empirically confirming our method’s efficacy.

我们引入了 PODS，一个轻量级框架，解决了现代 RLVR 训练中的一个基本瓶颈：在 embarrassingly parallel rollout 生成和内存密集型策略更新之间的不对称性。PODS 并行生成大量 rollouts，并仅根据最大方差规则选择一个信息性子集来更新策略。我们的分析表明，最优子集可以在 $O(n \log n)$ 时间内找到。这种简单而有原则的方法在相同的时间预算下始终优于标准的 GRPO，至少带来 $1.7\times$ 的加速效果，并在多种模型架构、规模和部署场景中达到更高的最终精度。我们的消融研究显示，只要 m 不太小，PODS 的性能在广泛的下采样比率范围内都具有鲁棒性，实证验证了我们方法的有效性。

Limitations. As we emphasize in the paper, our evaluation focuses on RLVR tasks where response correctness is verifiable. Other domains such as open-ended dialogue may exhibit distinct dynamics of the

限制。正如我们在论文中所强调的，我们的评估关注的是响应正确性可验证的 RLVR 任务。其他领域，如开放式对话，可能会表现出不同的动态特性。

algorithms. Moreover, in workloads that demand greater prompt diversity, similar gains might be obtained by processing more prompts per iteration with fewer rollouts per prompt and accumulating gradients across prompts, which is an alternative path to address the inference-update asymmetry. Finally, because PODS alters the training rollout distribution through selective down-sampling, it behaves off-policy and may be unsuitable when strict on-policy guarantees are required despite its improved empirical performance.

算法。此外，在需要更高提示多样性的工作负载中，通过每次迭代处理更多的提示、每个提示使用更少的 rollout，并在提示之间累积梯度，可能获得类似的收益，这是一种解决推理-更新不对称性的替代方法。最后，由于 PODS 通过选择性下采样改变了训练 rollout 分布，因此其行为属于 off-policy，尽管其在实证性能上有所改进，但在需要严格 on-policy 保证的情况下可能并不适用。

Scope of claims. Our empirical and theoretical claims in this paper are specific to PODS instantiated within GRPO. While we expect the inference-update asymmetry to be present in other RLVR pipelines, demonstrating that down-sampling yields comparable benefits for other RL methods (e.g., PPO-style objectives or value-based variants) requires separate analysis and experimentation.

主张的范围。本文中的经验性和理论性主张特定于在 GRPO 中实例化的 PODS。虽然我们预计推理-更新不对称性在其他 RLVR 流程中也存在，但要证明下采样对其他 RL 方法（例如 PPO 风格的目标或基于价值的变体）产生可比较的益处，需要进行单独的分析和实验。

Discussion and future work. PODS is a general framework that admits different down-sampling rules. In this paper, we focus on the max-variance rule due to its superior empirical performance in common RLVR settings. In some scenarios, other rules might be more effective. For example, when different prompts cause the model to have highly varying reward distributions, applying the max-variance rule across all rollouts may lead to over-sampling from a small subset of prompts with extreme difficulty levels. In such cases, applying the max-variance rule within each prompt and then selecting a balanced subset across prompts may be more effective. More

broadly, the down-sampling rule can be designed to take into account more information beyond the reward values, such as the rollouts' entropy, similarity measures between rollouts, or even a target reward distribution that we wish to down-sample towards. Exploring these alternative down-sampling rules and their empirical performance across different settings is an interesting direction for future work.

讨论与未来工作。PODS 是一个通用框架，它允许采用不同的下采样规则。在本文中，我们重点关注最大方差规则，因为它在常见的 RLVR 设置中表现出色。在某些场景下，其他规则可能更有效。例如，当不同的提示导致模型具有高度变化的奖励分布时，对所有 rollout 应用最大方差规则可能会导致对一小部分具有极端难度水平的提示进行过度采样。在这种情况下，对每个提示分别应用最大方差规则，然后在提示之间选择一个平衡的子集，可能会更有效。更广泛地说，下采样规则可以设计为考虑超出奖励值之外的更多信息，例如 rollout 的熵、rollout 之间的相似性度量，甚至是我们希望下采样以达到的目标奖励分布。探索这些替代下采样规则及其在不同设置下的实证表现，是未来工作的有趣方向。

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